

Equality and calm in the mind and vital parts, an intense psychic emotion in the heart can perfectly go together.

Sept 6
Sunday

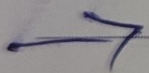
Cryptology - Theory.

Assignment - 2.

Encrypt / Decrypt using Ping Pong Paper.

①. Encrypt "CHASE YOUR DREAMS" using

- Caesar cipher;
- Shift cipher with key 7.
- ROT 13.



0	1	2	3	4	5	6	7	8	9	10
A	B	C	D	E	F	G	H	I	J	K
11	12	13	14	15	16	17	18	19		
L	M	N	O	P	Q	R	S	T		
20	21	22	23	24	25					
U	V	W	X	Y	Z					



Caesar Cipher

$$C = E(P, k).$$

Emotion is necessary in the Yoga and it is only the excessive emotional sensitiveness which makes one enter into despondency over small things that has to be overcome.

Sept 7
Monday

$$C = E(P + n) \bmod 26$$

$$n = 3.$$

$$C = E(P + n) \bmod 26.$$

$$C = (2 + 3) = 5 = F$$

$$H = (7 + 3) = 10 = K$$

$$A = (0 + 3) = 3 = D$$

$$S = (18 + 3) = 21 = V$$

$$E = (4 + 3) = 7 = H$$

$$Y = (24 + 3) = 27 \equiv 1 = A$$

$$O = (14 + 3) = 17 = R$$

$$U = (20 + 3) = 23 = X$$

$$R = (17 + 3) = 20 = U$$

$$D = (3 + 3) = 6 = G$$

$$K = (17 + 3) = 20 = U$$

$$E = (4 + 3) = 7 = H$$

$$A = (0 + 3) = 3 = D$$

$$M = (12 + 3) = 15 = P$$

$$S = (18 + 3) = 21 = V$$

The very basis
there can be no

Sept 8
Tuesday

①

C =

H =

A =

S =

E =

Y =

O =

U =

R =

D =

K =

E =

A =

M =

S =

Cipher

at sensi-
t has to

The very basis of this Yoga is bhakti and if one kills one's emotional being
there can be no bhakti.

Sept 8
Tuesday

RIPER TEXT:

"FKDVH ARXU QVHOPV"

① Shift cipher with $\rightarrow 7$.

$$C = E(P + 7) \bmod 26.$$

$$C = (2 + 7) \bmod 26 = 9 \Rightarrow J$$

$$H = (7 + 7) \bmod 26 = 14 \Rightarrow O$$

$$A = (0 + 7) \bmod 26 = 7 \Rightarrow H$$

$$S = (18 + 7) \bmod 26 = 25 \Rightarrow Z$$

$$E = (4 + 7) \bmod 26 = 11 \Rightarrow L$$

$$Y = (24 + 7) \bmod 26 = 5 \Rightarrow F$$

$$O = (14 + 7) \bmod 26 = 21 \Rightarrow V$$

$$U = (20 + 7) \bmod 26 = 1 \Rightarrow B$$

$$R = (17 + 7) \bmod 26 = 23 \Rightarrow X$$

$$D = (3 + 7) \bmod 26 = 10 \Rightarrow K$$

$$R = (17 + 7) \bmod 26 = 23 \Rightarrow X$$

$$E = (4 + 7) \bmod 26 = 11 \Rightarrow L$$

$$A = (0 + 7) \bmod 26 = 7 \Rightarrow H$$

$$M = (12 + 7) \bmod 26 = 19 \Rightarrow T$$

$$S = (18 + 7) \bmod 26 = 25 \Rightarrow Z$$

Plaintext: J O H Z L F V B X K Y L H U Z.

... what has to be done is not to stay on the surface but go deep within. For the psychic is there behind the emotional surface, deep behind the heart centre.

Once one reaches the untroubled

Sept 9 (c) ROT 13.
Wednesday

$$C = (P + 13) \bmod 26.$$

$$C = (2 + 13) \bmod 26 = 15 \quad - \quad P$$

$$H = (7 + 13) \bmod 26 = 20 \quad - \quad U$$

$$A = (0 + 13) \bmod 26 = 13 \quad - \quad N$$

$$S = (18 + 13) \bmod 26 = 5 \quad - \quad F$$

$$E = (4 + 13) \bmod 26 = 17 \quad - \quad R$$

$$Y = (24 + 13) \bmod 26 = 11 \quad - \quad L$$

$$O = (14 + 13) \bmod 26 = 1 \quad - \quad B$$

$$U = (20 + 13) \bmod 26 = 7 \quad - \quad H$$

$$R = (17 + 13) \bmod 26 = 4 \quad - \quad F$$

$$D = (15 + 13) \bmod 26 = 2 \quad - \quad C$$

$$R = (17 + 13) \bmod 26 = 4 \quad - \quad F$$

$$E = (4 + 13) \bmod 26 = 17 \quad - \quad R$$

$$A = (5 + 13) \bmod 26 = 18 \quad - \quad S$$

$$M = (12 + 13) \bmod 26 = 25 \quad - \quad Z$$

$$S = (18 + 13) \bmod 26 = 5 \quad - \quad F$$

Ciphertext: PUNFR LBME CERSZF

Sept 10
Thursday

(2)

(a)

W

R

W

D

O

O

B

V

H

F

U

M

W

Sept 10
Thursday

② Decrypt

⑤ WRWDOOB VHFUHW

PHVVDTJH" using Caesar cipher.

$$P = D(C - n) \bmod 26, \quad n = 3$$

$$W = (22 - 3) \bmod 26 = 19$$

$$R = (17 - 3) \bmod 26 = 14$$

$$W = (22 - 3) \bmod 26 = 19$$

$$D = (3 - 3) \bmod 26 = 0$$

$$O = (14 - 3) \bmod 26 = 11$$

$$O = (14 - 3) \bmod 26 = 11$$

$$B = (1 - 3) \bmod 26 = 24$$

$$V = (21 - 3) \bmod 26 = 18$$

$$H = (7 - 3) \bmod 26 = 4$$

$$F = (5 - 3) \bmod 26 = 2$$

$$U = (20 - 3) \bmod 26 = 17$$

$$H = (7 - 3) \bmod 26 = 4$$

$$W = (22 - 3) \bmod 26 = 19$$

There should be even in deep feeling a calm, a control, a purifying restraint and measure.

Sept 11 $D = (15-3) \text{ mod } 26 = M$

Friday $H = (7-3) \text{ mod } 26 = E$

$V = (21-3) \text{ mod } 26 = S$

$V = (21-3) \text{ mod } 26 = S$

$D = (3-3) \text{ mod } 26 = E$

$T = (9-3) \text{ mod } 26 = E$

$H = (7-3) \text{ mod } 26 = E$

Plain text: Totally secret Message

② $S T B D T W P Y N R J$

$S = (18-5) \text{ mod } 26 = N$

$T = (9-5) \text{ mod } 26 = E$

$B = (1-5) \text{ mod } 26 = W$

$D = (3-5) \text{ mod } 26 = Y$

$T = (19-5) \text{ mod } 26 = O$

$W = (22-5) \text{ mod } 26 = R$

$P = (15-5) \text{ mod } 26 = K$

$Y = (24-5) \text{ mod } 26 = T$

$N = (13-5) \text{ mod } 26 = I$

$R = (17-5) \text{ mod } 26 = M$

$J = (9-5) \text{ mod } 26 = E$

Plain text: New York Ann.

One should not be at the mercy of one's feelings and sentiments, but master of oneself always.

Sept 12
Saturday

(C)

FRAQ ZBER PBSSRR

using ROT 13.

$$P = (C - 13) \bmod 26.$$

$$F = (5 - 13) \bmod 26 = 5$$

$$R = (17 - 13) \bmod 26 = 4$$

$$A = (0 - 13) \bmod 26 = 13$$

$$Q = (16 - 13) \bmod 26 = 3$$

$$Z = (25 - 13) \bmod 26 = 12$$

$$B = (1 - 13) \bmod 26 = 14$$

$$E = (4 - 13) \bmod 26 = 15$$

$$R = (17 - 13) \bmod 26 = 4$$

$$P = (15 - 13) \bmod 26 = 2$$

$$B = (1 - 13) \bmod 26 = 14$$

$$S = (18 - 13) \bmod 26 = 5$$

$$S = (18 - 13) \bmod 26 = 5$$

$$R = (17 - 13) \bmod 26 = 4$$

$$R = (17 - 13) \bmod 26 = 4$$

Plaintext: Send more coffee.